# Supporting Information for : Continent-scale macroevolutionary insights from the last 23 million years of South American mammalian diversity

Lucas Buffan[1], Myriam Boivin[2], Narla S. Stutz[1], François Pujos[3], Pierre-Olivier Antoine[1], Fabien L. Condamine[1], Laurent Marivaux[1]

Affiliations:
[1] Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France.
[2] Instituto de Ecorregiones Andinas (INECOA), Universidad Nacional de Jujuy, CONICET, IdGyM, Avenida Bolivia 1661, 4600 San Salvador de Jujuy, Jujuy, Argentina.
[3] IANIGLA, CCT–CONICET–Mendoza, Avenida Ruiz Leal s/n, Parque General San Martín, 5500 Mendoza, Argentina.

This file contains:

- Supplementary text
- Supplementary figures
- Supplementary tables

Other Supplementary Materials may be found at `<placeholder>`.
Codes are available at: https://github.com/Buffan3369/PostPal_SAM.git

# Supplementary text

## About occurrence data cleaning

As done in Buffan et al. (in press), we proceeded to an intense cleaning of the genus-level occurrence primarily downloaded from *The Paleobiology Database*. Occurrences of uncertain affinity, presenting elements of open nomenclature such as *cf.*, *aff.* or *?* were removed from the database. In addition, taxa associated to an inappropriate paper (*i.e.*, that made no clear mention of it) were removed from the database. Below, we present the great lines of our cleaning procedure, taxonomic subdivision per taxonomic subdivision. For tracability reasons, each occurrence that was not linked to a systematics paper describing it – or clearly referring to a such a paper – were removed from our dataset.

### Caviomorpha

Age ranges of the Miocene occurrences from the Río Santa Cruz (Santacrucian SALMA) were sharpened based on their assigned locality (Arnal et al. 2019). Hence, occurrence from Barrancas Blancas were respectively assigned the 16.3–17.21 and 15.3–16.47 Ma intervals.

Given that *Lagostomus (Lagostomopsis)* is a subgenus (Rasia and Candela 2012), we set the `genus` column of each *Lagostomus (Lagostomopsis)* occurrence to *Lagostomus*, and switched their `genus_level_status` from `extinct` to `extant`.

Uncertain age of the Ituzáingo formation was set to Huayquerian (Late Miocene), *ca.* 5–8 Ma.

The single occurrence of *Colpostemma sinuata* (Ameghino, 1891) was renamed *Miocastor sinuata* according to the synonymisation established by Candela and Noriega (2004).

All occurrences of the genus *Kiyutherium* were synonymised with *Cardiatherium*, following Vucetich et al. (2012).

Age range of occurrences from Bucher et al. (2021) were set to the one of the original publication: 14.86–12.75 Ma when coming from the "Cruces Infinitos" and "Los Yeguarizos localities", 12.75–12 Ma when coming from the "La Gloria" and "La Hoyada" localities.

Based on Brandoni et al. (2016), the age range of the occurrences from the Pinturas fauna (*e.g.*, (Kramarz 2002; Kramarz 2004; Kramarz 2006)), Burdigalian, were set between 20 and 16.8 Ma.

According to (Esteban et al. 2014; Esteban et al. 2019), the age range of the occurrences belonging to the Andalhualá formation was set to Huayquerian–Chapadmalalan (*c.* 7.14–3.66 Ma).

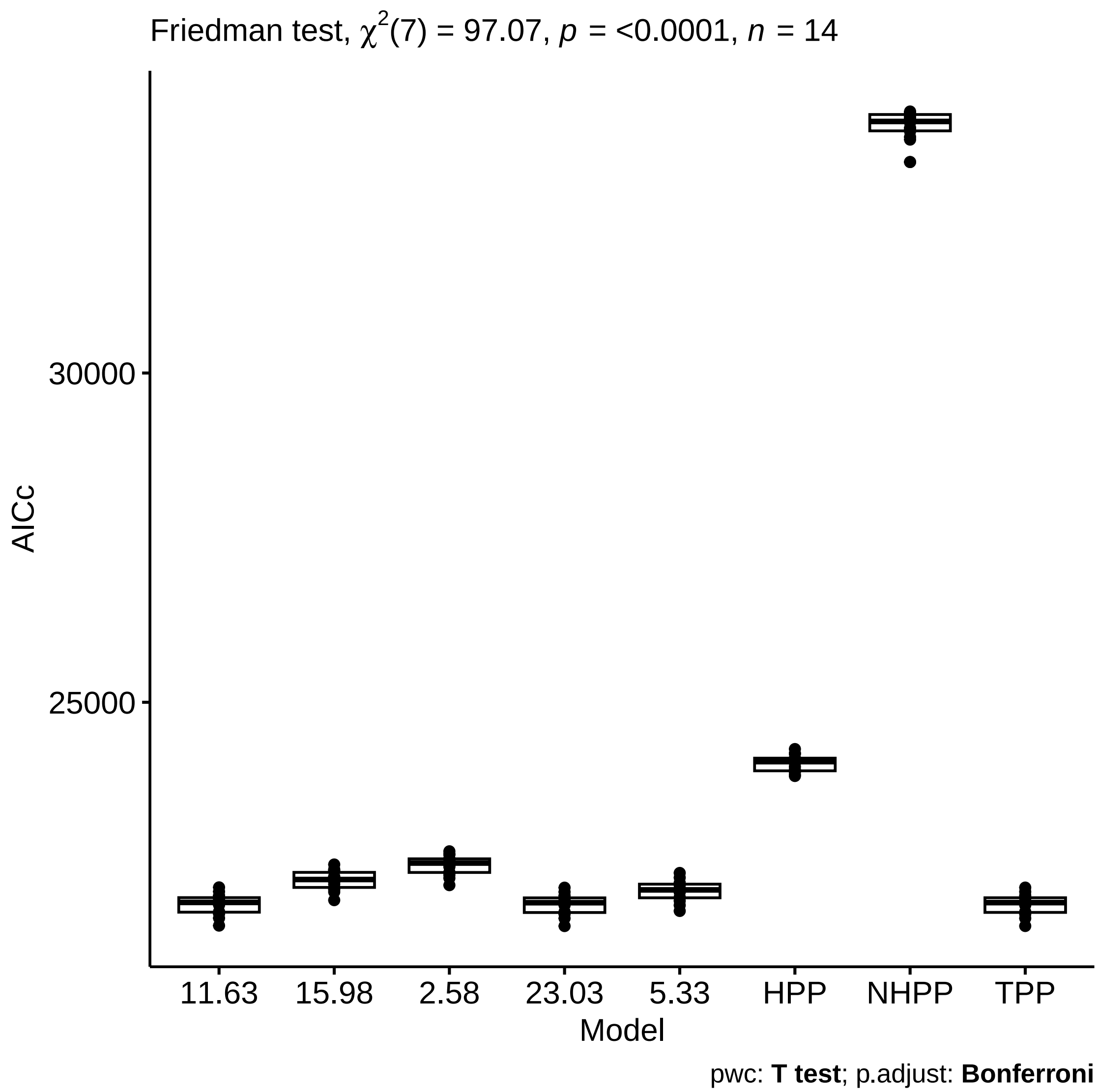
*Perimys perpinguis* and *Perimys impactus* lumped in *Perimys onostus* according to McGrath et al. (2022).

Age range of the occurrence belonging to the Sierra Baguales (Santa Cruz formation, Magallanes, Chile) were set to 19–17.8 Ma according to the isotopic constraints provided in Bostelmann et al. (2013).

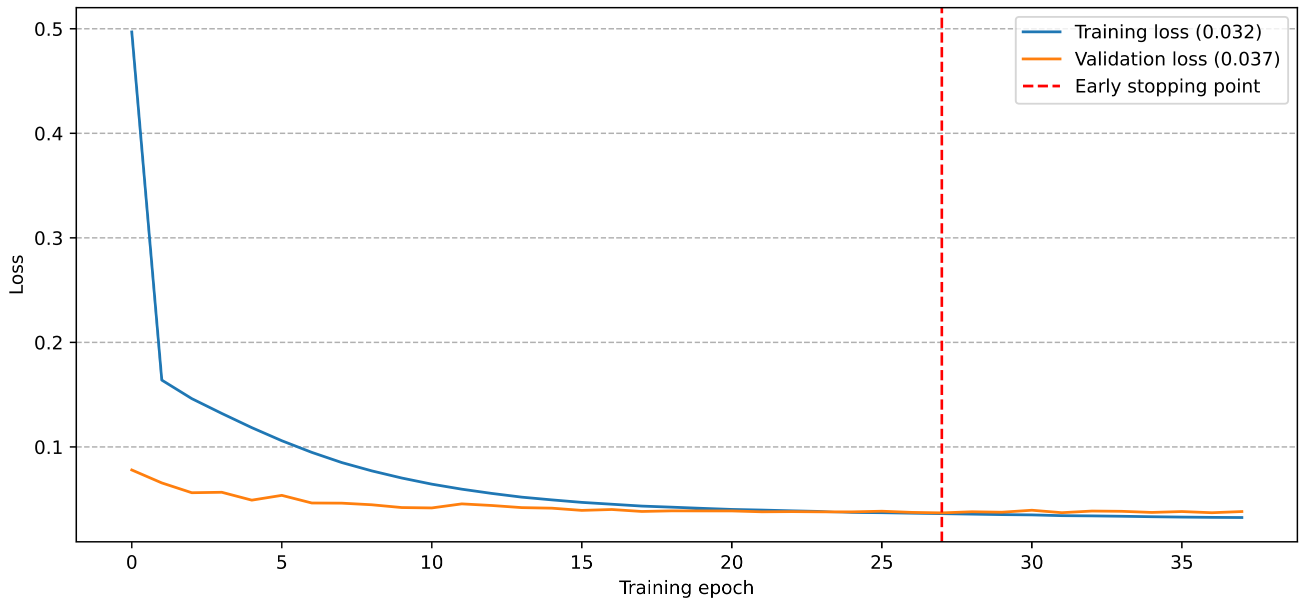
Following the PhD Thesis of María Encarnación Pérez Pérez (2010), we synonimised each occurrence of *Eocardia perforata* and *Eocardia divisa* with *Eocardia montana*. **Precise other synonymies**. Also, we directly sourced this thesis to complete occurrence from the fossil record of the family Eocardiidae.

Age range of the occurrences reported from the Cerro Boleadoras Formation was set to 16.5–15.1 Ma, according to Vizcaino et al. (2022).

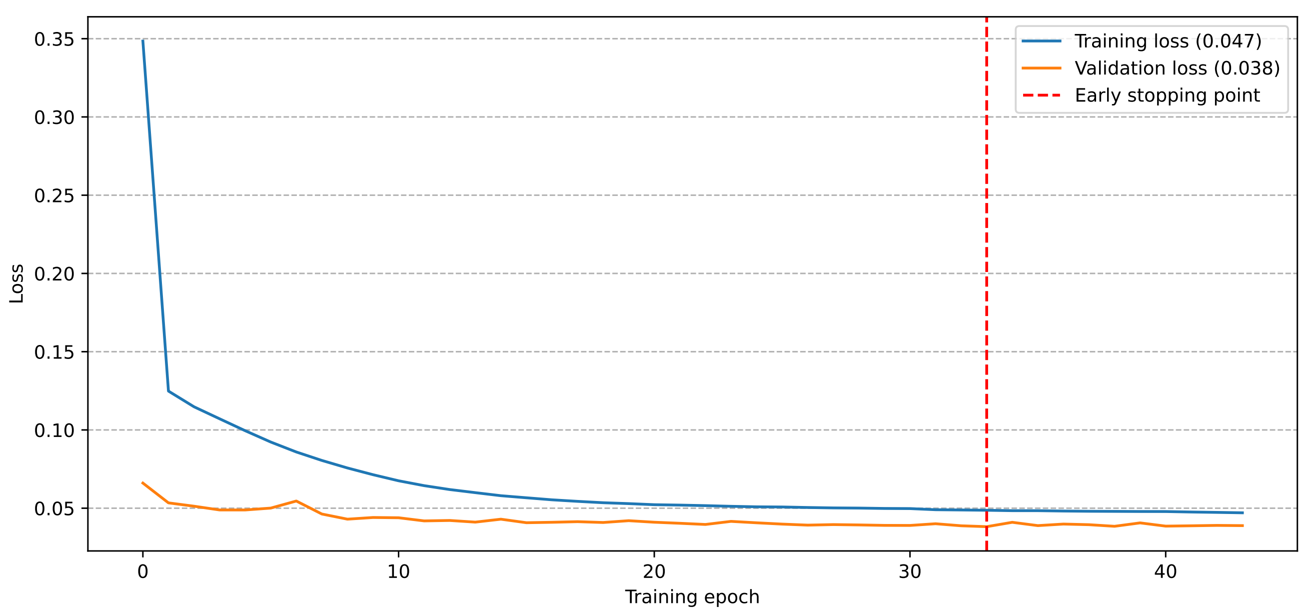
## Supplementary Figures




**Supplementary Figure 1.** Selection of the preservation model best fitting our occurrence data, based on a maximum likelihood test incorporated in PyRate (Silvestro et al. 2019). HPP, NHPP and TPP respectively stand for Homogeneous, Non-Homogeneous and Time-variable Poisson Processes for modelling preservation in our fossil data. In the case of TPP, we allowed for preservation rate to vary across geological stages (with only one bin for Quaternary). Models on the left of the x axis indexed by numbers are TPP with one removed bin, the lower (*i.e.*, older) boundary of which corresponds to the aforementioned number. Best model: TPP without the 23.03−15.98 Ma bin (mean $AICc = 21928.78$, **Supp. tab. 1**).




**Supplementary Figure 2.** Training and validation loss across training epochs for a DeepDive network comprising 3 Long Short Term Memory (LSTM) units (128, 64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the "early stopping point", *i.e.*, when the validation loss starts icreasing. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.




**Supplementary Figure 3.** Training and validation loss across training epochs for a DeepDive network comprising 2 LSTM units (64 and 32 nodes) connected to one dense layer (32 nodes) before the output layer. The red vertical line indicates the "early stopping point", *i.e.*, when the validation loss starts icreasing. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

# Supplementary Tables

**Supplementary Table 1.** Mean AICc assigned to each preservation model given our occurrence data, across the 20 replicates of age assignment. HPP, NHPP and TPP respectively stand for Homogeneous, Non-Homogeneous and Time-variable Poisson Processes for modelling preservation in our fossil data. In the case of TPP, we allowed for preservation rate to vary across geological stages (with only one bin for Quaternary). Models indexed by numbers are TPP with one removed bin, the lower (*i.e.*, older) boundary of which corresponds to the aforementioned number. The model best fitting our data (*i.e.*, having the lowest AICc) is shown in bold.

| | HPP | NHPP | TPP | 23.03 | 15.98 | 11.63 | 5.33 | 2.58 |
|---|---|---|---|---|---|---|---|---|
| mean AICc | 24074.36 | 33759.41 | 21930.08 | **21928.78** | 22296.44 | 21932.97 | 22137.15 | 22525.55 |

**Supplementary Table 2.** Posterior estimates of the multivariate birth-death correlation parameters with the origination rate ($G_\lambda$) for our full early Neogene (23.04–13.82 Ma) dataset, and their associated Shrinkage weights ($w_\lambda$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\lambda$ | $G_\lambda$ 95% HPD | $w_{G,\lambda}$ | $w_{G,\lambda}$ 95% HPD |
|---|---|---|---|---|
| Self-diversity | -0.905 | [-1.852; 0.031] | 0.609 | [0.013; 0.984] |
| Global Temperature | 0.333 | [-0.111; 0.96] | 0.631 | [0.003; 0.987] |
| **Andes Elevation** | **0.274** | **[0.1; 0.461]** | **0.884** | **[0.408; 0.995]** |
| **Sea Level** | **-0.017** | **[-0.038; 0]** | **0.778** | **[0.017; 0.993]** |

**Supplementary Table 3.** Posterior estimates of the multivariate birth-death correlation parameters with the extinction rate ($G_\mu$) for our full early Neogene (23.04–13.82 Ma) dataset, and their associated Shrinkage weights ($w_\mu$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\mu$ | $G_\mu$ 95% HPD | $w_{G,\mu}$ | $w_{G,\mu}$ 95% HPD |
|---|---|---|---|---|
| **Self-diversity** | **-1.521** | **[-2.466; -0.647]** | **0.735** | **[0.217; 0.986]** |
| Global Temperature | 0.187 | [-0.197; 0.697] | 0.537 | [0.002; 0.984] |
| Andes Elevation | 0.001 | [-0.102; 0.115] | 0.249 | [0; 0.962] |
| **Sea Level** | **0.007** | **[-0.005; 0.022]** | **0.542** | **[0.004; 0.981]** |

**Supplementary Table 4.** Posterior estimates of the multivariate birth-death correlation parameters with the origination rate ($G_\lambda$) for our full late Neogene (13.82–2.58 Ma) dataset, and their associated Shrinkage weights ($w_\lambda$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\lambda$ | $G_\lambda$ 95% HPD | $w_{G,\lambda}$ | $w_{G,\lambda}$ 95% HPD |
|---|---|---|---|---|
| Self-diversity | -0.731 | [-1.506; 0.021] | 0.482 | [0.01; 0.981] |
| Global Temperature | 0.211 | [-0.089; 0.538] | 0.513 | [0.004; 0.978] |
| Andes Elevation | 0.042 | [-0.031; 0.138] | 0.385 | [0.001; 0.973] |
| **Sea Level** | **-0.024** | **[-0.041; -0.002]** | **0.656** | **[0.049; 0.988]** |

**Supplementary Table 5.** Posterior estimates of the multivariate birth-death correlation parameters with the extinction rate ($G_\mu$) for our full early Neogene (13.82–2.58 Ma) dataset, and their associated Shrinkage weights ($w_\mu$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\mu$ | $G_\mu$ 95% HPD | $w_{G,\mu}$ | $w_{G,\mu}$ 95% HPD |
|---|---|---|---|---|
| **Self-diversity** | **-1.24** | **[-2; -0.523]** | **0.669** | **[0.167; 0.986]** |
| **Global Temperature** | **-0.4** | **[-0.717; -0.07]** | **0.714** | **[0.097; 0.99]** |
| **Andes Elevation** | **-0.128** | **[-0.224; -0.03]** | **0.692** | **[0.124; 0.987]** |
| Sea Level | 0.002 | [-0.006; 0.017] | 0.172 | [0; 0.958] |

**Supplementary Table 6.** Posterior estimates of the multivariate birth-death correlation parameters with the origination rate ($G_\lambda$) for our full Quaternary (2.58–0 Ma) dataset, and their associated Shrinkage weights ($w_\lambda$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\lambda$ | $G_\lambda$ 95% HPD | $w_{G,\lambda}$ | $w_{G,\lambda}$ 95% HPD |
|---|---|---|---|---|
| Self-diversity | 0.015 | [-0.588; 0.743] | 0.161 | [0; 0.945] |
| **Global Temperature** | **0.725** | **[0.484; 0.969]** | **0.762** | **[0.307; 0.991]** |
| Andes Elevation | -0.018 | [-0.218; 0.139] | 0.11 | [0; 0.941] |
| **Sea Level** | **0.027** | **[0.014; 0.046]** | **0.579** | **[0.133; 0.986]** |

**Supplementary Table 7.** Posterior estimates of the multivariate birth-death correlation parameters with the extinction rate ($G_\mu$) for our full Quaternary (2.58–0 Ma) dataset, and their associated Shrinkage weights ($w_\mu$). Median estimates are shown along with their 95% Highest Posterior Density (HPD) interval. Significant correlations, *i.e.*, having a $w_\lambda$ larger than 0.5 and of which the 95% HPD of the associated coefficients do not overlap 0 are shown in bold.

| Covariate | $G_\mu$ | $G_\mu$ 95% HPD | $w_{G,\mu}$ | $w_{G,\mu}$ 95% HPD |
|---|---|---|---|---|
| **Self-diversity** | **4.081** | **[2.553; 5.944]** | **0.934** | **[0.678; 0.998]** |
| Global Temperature | -0.016 | [-0.47; 0.294] | 0.188 | [0; 0.955] |
| **Andes Elevation** | **-0.99** | **[-2.424; 0]** | **0.864** | **[0.021; 0.997]** |
| Sea Level | -0.003 | [-0.047; 0.022] | 0.27 | [0; 0.967] |

**Supplementary Table 8.** Performance of the different network architectures tested in our DeepDive analyses. `LSTM Layers` and `Fully-connected` are respectively the number of Long Short Term Memory and Fully-connected units of the network (3: 128, 64 and 32 nodes; 2: 64 and 32 nodes; 1: 32 nodes). `Dropout` represents the Monte-Carlo dropout fraction, that is, the proportion of connections between neurons randomly toggled off between two learning epochs – one epoch being characterised by a complete pass through the entire training dataset –, preventing from overfitting. `Nb. training epochs` is the optimal number of training epochs. `Training loss` and `Validation loss` are the values of the Mean Squared Error (MSE) estimated between the 'true' diversity of each simulated dataset (resp. from the training or the validation set), and the predicted diversity made by the network after $n$ rounds of optimisation, where $n$ is the optimal number of training epochs. `Evaluation loss` represent the MSE between each true and its associated predicted diversity trajectory from the test datset, once the is trained (*i.e.*, after $n$ rounds of optimisation, where $n$ is the optimal number of training epochs). The optimal number of epochs is determined when the `Validation loss` starts increasing (see *Supp. Fig. 1 and 2*). Architectures resulting in the smallest evaluation loss values, *i.e.*, which we believe to generate the most accurate predictions, are depicted in bold.

| LSTM layers | Fully-connected | Dropout | Nb. training epochs | Training loss | Validation loss | Evaluation loss |
|---|---|---|---|---|---|---|
| **3** | **2** | **0.05** | **27** | **0.032** | **0.037** | **0.039** |
| 3 | 1 | 0.05 | 17 | 0.037 | 0.038 | 0.04 |
| 3 | 0 | 0.05 | 37 | 2.511 | 2.514 | 2.529 |
| 2 | 2 | 0.05 | 23 | 0.038 | 0.038 | 0.04 |
| **2** | **1** | **0.05** | **33** | **0.047** | **0.038** | **0.039** |
| 2 | 0 | 0.05 | 18 | 2.511 | 2.514 | 2.529 |
| 1 | 1 | 0.05 | 64 | 0.045 | 0.04 | 0.041 |
| 1 | 0 | 0.05 | 14 | 2.511 | 2.514 | 2.529 |

## Supplementary references

Arnal M., Pérez M.E., Deschamps C. 2019. Revision of the Miocene Caviomorph Rodents From the Río Santa Cruz (Argentinean Patagonia). Publicación Electrónica de la Asociación Paleontológica Argentina. 19.

Bostelmann J.E., Roux J.P.L., Vásquez A., Gutiérrez N.M., Oyarzún J.L., Carreño C., Torres T., Otero R., Llanos A., Fanning C.M., Hervé F. 2013. Burdigalian deposits of the Santa Cruz Formation in the Sierra Baguales, Austral (Magallanes) Basin: Age, depositional environment and vertebrate fossils. Andean Geology. 40:458–489.

Brandoni D., Ruiz L.G., Tejedor M.F., Martin G., Fleagle J.G. 2016. Megatherioidea (Mammalia, Xenarthra, Tardigrada) from the Pinturas Formation (Early Miocene), Santa Cruz Province (Argentina) and their chronological implications. PalZ. 90:619–628.

Bucher J., Pérez M.E., González Ruiz L.R., D'Elía L., Bilmes A. 2021. New middle miocene (langhian - serravallian) vertebrate localities in northwestern patagonia, argentina: A contribution to high latitude south american land mammal ages sequence. Journal of South American Earth Sciences. 107:103024.

Candela A.M., Noriega J.I. 2004. Los coipos (rodentia, caviomorpha, myocastoridae) del "mesopotamiense" (mioceno tardío; formación ituzaingó) de la provincia de entre ríos, argentina. Temas de la Biodiversidad del Litoral fluvial argentino II.:77–82.

Esteban G.I., Georgieff S.M., Nasif N.L., Ibáñez L., Bonini R.A. 2019. Marco estratigráfico, paleontología de vertebrados y paleoambientes del plioceno de catamarca, tucumán y santiago del estero, argentina. p. 348–366.

Esteban G.I., Nasif N.L., Georgieff S.M. 2014. Cronobioestratigrafia del mioceno tardio-plioceno temprano, puerta de corral quemado y villavil, provincia de catamarca, argentina. Acta geológica lilloan. 26:165–192.

Kramarz A. 2002. Roedores chinchilloideos (Hystricognathi) de la Formación Pinturas, Mioceno temprano-medio de la provincia de Santa Cruz, Argentina. Revista del Museo Argentino de Ciencias Naturales nueva serie. 4:167–180.

Kramarz A.G. 2004. Octodontoids and erethizontoids (Rodentia, Hystricognathi) from the Pinturas Formation, Early - Middle Miocene of Patagonia, Argentina. Ameghiniana. 41:199–216.

Kramarz A.G. 2006. Eocardiids (rodentia, hystricognathi) from the pinturas formation, late early miocene of patagonia, argentina. Journal of Vertebrate Paleontology. 26:770–778.

McGrath A.J., Chick J., Croft D.A., Dodson H.E., Flynn J.J., Wyss A.R. 2022. Cavioids, chinchilloids, and erethizontoids (hystricognathi, rodentia, mammalia) of the early miocene pampa castillo fauna, chile. American Museum Novitates. 2022:1–46.

Pérez M.E. 2010. Sistemática, ecología y bioestratigrafía de Eocardiidae (Rodentia: hystricognathi: cavioidea) del Mioceno temprano y medio de Patagonia.

Rasia L.L., Candela A.M. 2012. Systematic and biostratigraphic significance of a chinchillid rodent from the pliocene of eastern argentina. Acta Palaeontologica Polonica. 58:241–254.

Silvestro D., Salamin N., Antonelli A., Meyer X. 2019. Improved estimation of macroevolutionary rates from fossil data using a Bayesian framework. Paleobiology. 45:546–570.

Vizcaino S.F., Bargo M.S., Pérez M.E., Aramendía I., Cuitiño J.I., Monsalvo E.S., Vlachos E., Noriega J.I., Kay R.F. 2022. Fossil vertebrates of the early-middle Miocene Cerro Boleadoras Formation, northwestern Santa Cruz Province, Patagonia, Argentina. Andean Geology. 49:382–422.

Vucetich M.G., Deschamps C.M., Vieytes E.C., Montalvo C.I. 2012. Late miocene capybaras from argentina: Skull anatomy, taxonomy, evolution, and biochronology. Acta Palaeontologica Polonica. 59:517–535.